Wildchrokie ms

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Methods

Common garden setup and data collection

We collected seeds from 18 species of deciduous trees and shrubs from four provenances across a 3.5° latitudinal gradient (Figure 1 and Table S1) in 2014-2015, germinated them following standard germination protocols and grew them to seedlings (Baskin & Baskin, 2014). In the spring of 2017, we planted them outside in the garden located near the Weld Hill Research Building at the Arnold Arboretum in Boston MA (42.30°N, 71.13°W). Plots were regularly weeded and watered throughout the duration of the study and were pruned in the fall of 2020 (Buonuito, unpublished).

For the years 2018-2020, we monitored leaf phenology for all individuals in the common garden once or twice per week from February to December; the exact monitoring frequency varies (see Figure S1). For this, we used a modified BBCH scale to observe the phenological stages (e.g. leafout: BBCH 15 and used an additional phase for budset: BBCH 102; *REF). We had a total of 239 leafout and 230 budset observations (which we refer as the full datasets), together yielding 227 full growing season observations (restricted dataset).

In the spring of 2023, we collected tree ring samples only for the tree species with high survivorship. Thus, we have four species in the *Betulaceae* family : Grey alder (*Alnus incana*), Yellow birch (*Betula alleghaniensis*), Paper birch (*Betula papyrifera*) and Grey birch (*Betula populifolia*). We collected cross-sections on 78 individuals by cutting the trunk above the root collar, and cores on 43 individuals using standard dendrochonological methods (Cook & Kairiukstis, 1990), for a total of 81 individuals (Table S1). We stored the cores in modified plastic straws to allow aeration and left them to dry with the cross-sections at ambient temperature for three months.

Sample processing, imaging and measuring

We mounted cores on wooden mounts, and sanded both the cores and cross-sections using progressively finer sandpaper grits (150, 300, 400, 600, 800, 1000), then scanned the cores and cross-sections at a resolution of 6250 dpi with a high-resolution treering scanner (Fong, unpublished). Using the digitized images, we measured ring widths with ImageJ (Schneider *et al.*, 2012) at three standard locations for each cross-section (at 120° from each other, averaging the three measurements to yield one ring width per year, per individual), and one ring width for each core. When we had cross-sections and cores for an individual, we used only the measurement from the cross sections ($n = 81$).

We performed visual cross dating. We did not perform statistical crossdating or detrending because our chronologies (years of data) were extremely short (Raden *et al.*, 2020) and showed no evidence of consistent allometric trends (Figure S2). We controlled for effects of different years by including year in our models (see 'Data analyses' below)

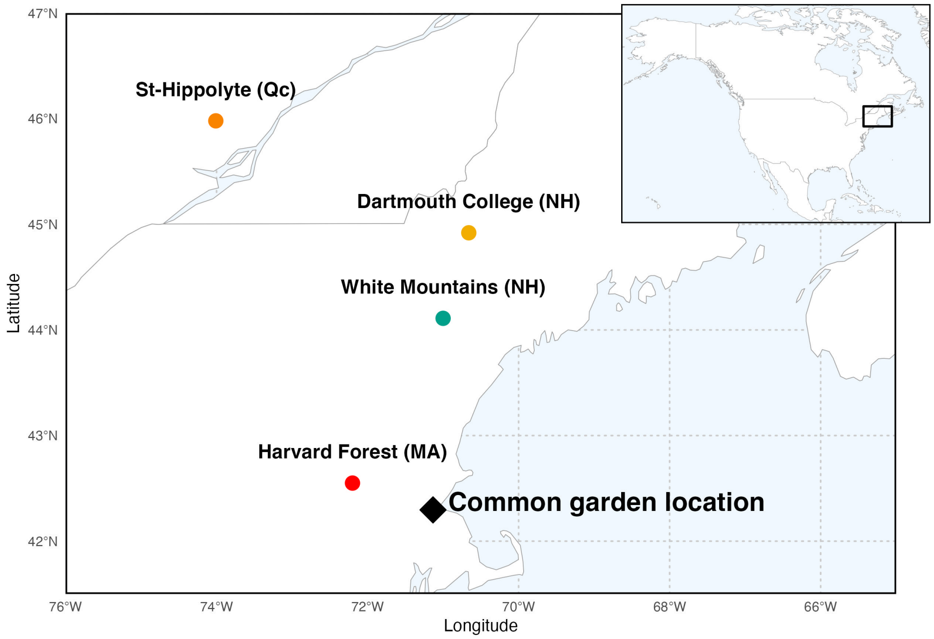

Figure 1: Provenance where the seeds were sourced from are depicted as circles, while the diamond denotes the location of the common garden. The common garden location is where the study took place at the Weld Research Building (Boston, MA). The colors of the dots match the model estimates of Figure 3.

Data analyses

To understand how growth shifted with season lengths we used linear Bayesian hierarchical models, estimating how growth, measured as ring width, covaried with season length, measured in four different season ways: GDD, GSL, SOS and EOS. We calculated the thermal season using growing degree days (GDD; mean daily temperature above 5°C) accumulated between leafout and budset, and the calendar season (or growing season length (GSL)) as the number of days between leafout and budset. Finally, we used leafout and budset as the start (SOS) and end of the season (EOS) as separate predictors that represent the boundaries of the growing season. For interpretability, we report ring width change as the averaged number of GDDs during 7 days in May across our three years —day of year 121 to 150 for the thermal season, and ring width change per seven days of season length, leafout and budset for GSL, SOS and EOS, respectively.

We used the daily climate data from the climate station located at Weldhill research station for 2018 and 2019 (Arboretum, 2026). Because the data was not available for that station after August 2020, we used the daily climate data from the weather station "Boston (Weld Hill)" of the Network for Environment and Weather Applications (NEWA) for the whole year in 2020 (**?**).

For each predictor, we estimated its effect on ring width observed (i, denoting each observation) for each tree ($treeid$) in each year ($year$) as:

$$
\begin{aligned}
ln(ringwidth)_i &\sim N(\mu, \sigma_y) \\
\mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{prov}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{species} \times \text{season predictor} \\
\alpha_{\text{prov}} &\sim N(0, \sigma_{\alpha\,\text{prov}}) \\
\alpha_{\text{treeid}} &\sim N(0, \sigma_{\alpha\,\text{treeid}})
\end{aligned}
$$

This model estimates a species-specific effect of each season length predictor (e.g. GDD) on growth ($\beta_{species}$), while also controlling for other variation due to species ($\alpha_{species}$), provenance (α_{prov}), individual tree (α_{treeid}) and year (α_{year}). We ran the models using both natural units and scaled predictors through z-score (REF?).

We used weakly informative priors for all models (increasing the priors by 2x for the z-scored models did not qualitatively affect the results). We ran four chains with 2000 warmup and 2000 sampling iterations each, for a total of 8000 sampling iterations. The models did not have any divergent transitions, had a $\hat{R}$ below 1.01, and values of effective sample size (n_{eff}) greater than 10% of the model iterations. We wrote our models using Stan (Carpenter *et al.*, 2017) with the rstan package version 2.32.7 (Stan Development Team, 2025) to run the Stan code in R.

Because ring widths decrease with age—particularly with such young trees—, we ran the models using basal area increment (BAI), as the response variable. This allows for a more accurate representation of annual increment by using the total ring area, instead of its radius. For this, we summed each ring width separately at the three distinct locations on the cross sections yielding three radii from pit to bark. We then calculated the area of each ring by subtracting the radius from the inner ring from the outer ring: $\pi \times (\text{Inner radius}^2 - \text{Outer radius}^2)$. and averaged across the three BAI for each ring. We also fit the models for SOS and EOS using both the restricted (n =227) and full datasets for each predictor (SOS n =239 and EOS n =230.

Results

Summary basic information

Our dataset includes four deciduous tree species from four sites with leaf phenology data spanning three years (2018 to 2020) for a total of 239 leafout and 230 budset observations, yielding 227 complete growing seasons (GS) for 81 individuals. At the common garden location, the mean summer temperature was similar across years with 2019 being the coolest (22.38°C), 2020 the hottest (23.07°C), and 2018 in between (22.85°C). There was no drought for those three years, except for in 2020, where a moderate summer drought ramped up to a severe drought in September.The calendar and thermal seasons ranged from 90 days and 1465.1 GDD (*B. alleghaniensis* in 2018) to 173 days and 2513.6 GDD (*A. incana* in 2018). The earliest leafout was 22-Apr (*A. incana* in 2019) and the latest, 05-Jun (also *A. incana*, but in 2020). The earliest budset was 13-Aug (*B. alleghaniensis* in 2018) and the latest, 24-Oct (*A. incana* in 2018). Finally, annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018; Figure S7).

Does tree growth increase with longer growing seasons?

Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it. Warmer thermal season increased growth for *B. papyrifera* (β = 0.44, UI_{90} [0.18, 0.7]), *B. populifolia* (β = 0.22, UI_{90} [0.07, 0.36]), and *B. alleghaniensis* (β = 0.13, UI_{90} [0.01, 0.24]), *A. incana* trended in the same direction (β = 0.07, UI_{90} [-0.04, 0.17]; Figure 2a and e; TableS2). Unlike its response to warmer thermal season, *A. incana* grew more with longer calendar seasons (β = 0.05, UI_{90} [0. 0.1]). *B. papyrifera* also grew more with longer calendar seasons (β = 0.18, UI_{90} [0.05, 0.32]), while the other two *Betula* did not, though they are trending in the same same way (*B. alleghaniensis*: β = 0.03, UI_{90} [-0.02, 0.09] and *B. populifolia*: β = 0.05, UI_{90} [-0.02, 0.13]; Figure 2b and f; Table S2).

On a scaled axis, both predictors were similarly predictive (Figure S10). Additionally, the results with basal area increment (BAI) for the GDD model were broadly similar to ring width (see Table S5).

Did a later start and end of season increase growth?

Increased growth with longer seasons is often presumed to come from growth in the spring, since spring phenology has shifted much more than fall phenology. However, while we found that a longer calendar season increased growth for two species, only one also grew more with an earlier start of season (*A. incana*: β = -0.07, UI_{90} [-0.14, -0.01]; Figure 2c and g; Table S2). However, *B. alleghaniensis* which did not grow more with longer calendar season, grew less with an earlier start of season (β = 0.1, UI_{90} [0, 0.2]; Figure 2c and g; Table S2)—but more with a later end of season (β = 0.12, UI_{90} [0.05, 0.18]). Alongside *B.*

alleghaniensis, two other species grew more with a later end of season (*B. populifolia*: $\beta = 0.1$, UI_{90} [0.02, 0.17], and *B. papyrifera*:$\beta = 0.21$, UI_{90} [0.09, 0.33]), with the latter being the only one of those three that also grew more with longer calendar season (Figure 2d and h; Table S2). The estimates were unchanged when using the full (SOS $n = 239$ and $n = 230$) or restricted datasets ($n = 227$; Figure S3a).

How did tree growth change across years, provenances and tree ids?

We did not find any clear effect of year on growth (2018: , [,], 2019: , [,] and 2020: , [,]. (Figure S7) and there was no apparent differences of growth looking across species (Figure S4 and S5). Growth did not change across any provenances (Figure 3; Table S3). The variation across provenances ($\sigma = 0.18$, UI_{90} [0.02, 0.5) and tree ids ($\sigma = 0.17$, UI_{90} [0.05, 0.27]) was similar.

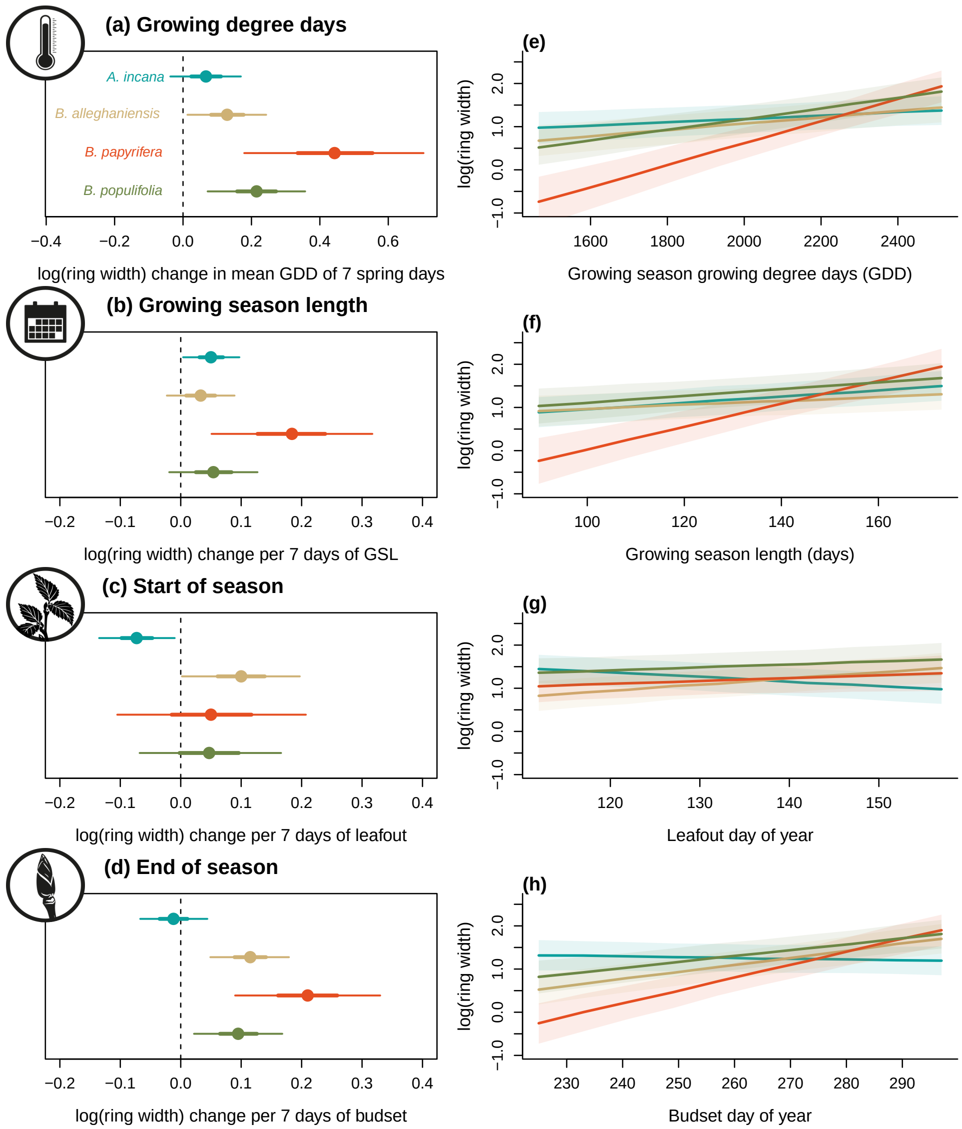

Figure 2: Estimates of ring width change (shown as ring width change per 7 average spring days GDD (174.05) as a function of four different predictors across four species. GDD represent the accumulated growing degree days between leafout and budset and the models estimates shows (a). GSL is the number of days between leafout and budset (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through budset (d). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

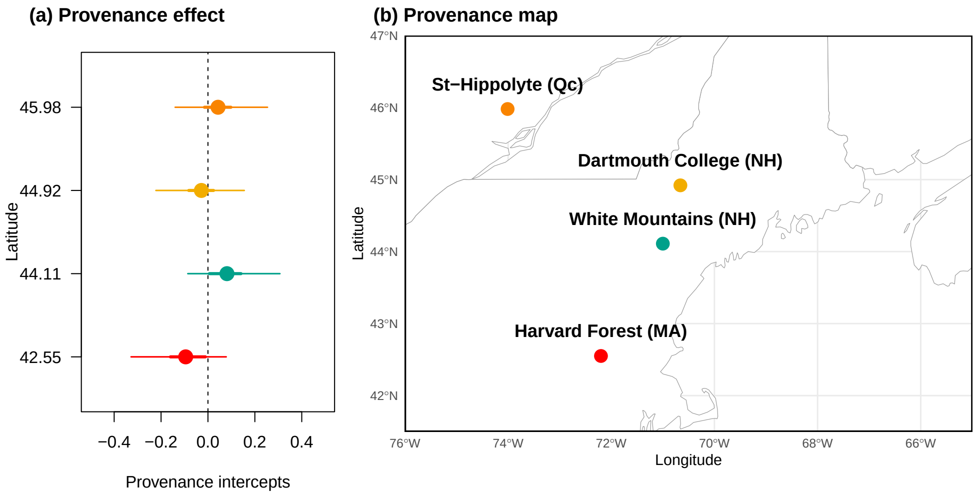

Figure 3: The effect of provenance on growth across a latitudinal gradient. (a) Model intercept parameters for each provenance with growing degree days (GDD) as the predictor (effects were similar for the other predictors, see Table S3). The dots represent the mean of each provenance and the thick lines, the 50% and 90% uncertainty intervals (UI). The sites are color-coded from (b), the map showing their locations.

Discussion

References

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Supplemental

Supplemental Methods

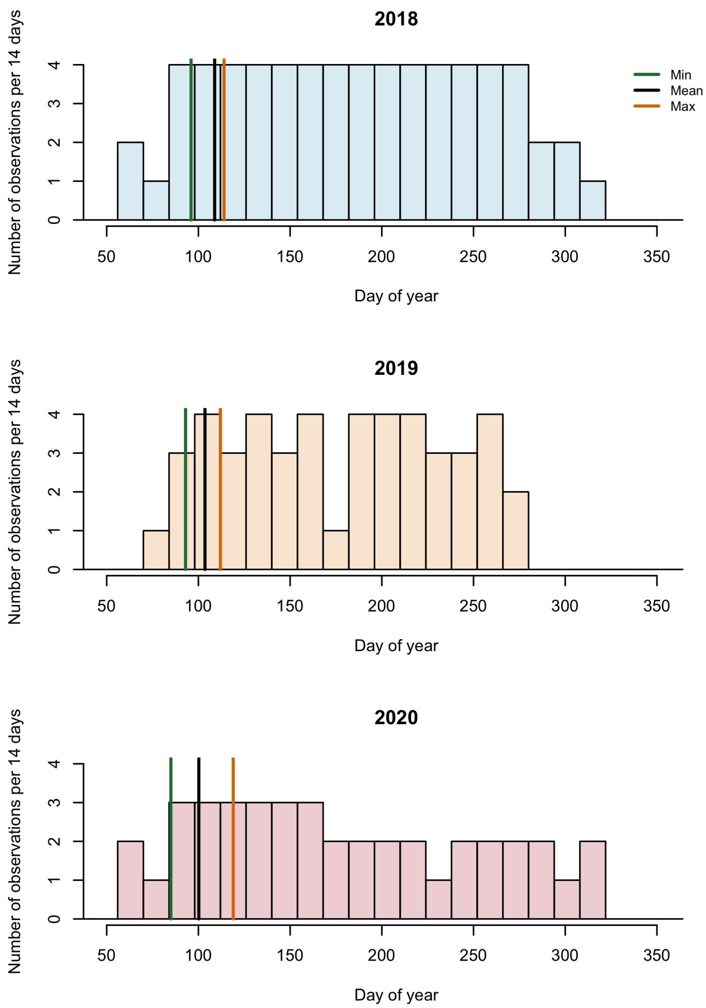

Figure S1: Phenology observatons bi-weekly.

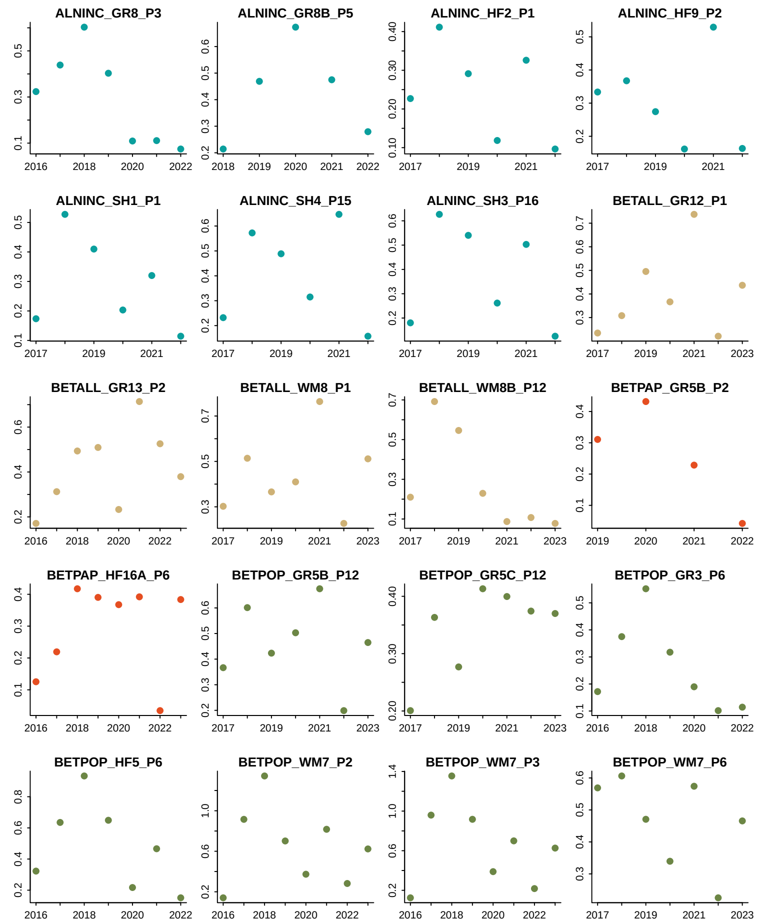

Figure S2: We randomly sampled 20 trees from our dataset and represent the ring width as a function of year to represent the allometry patterns across all years we had ring width for.

Table S1: Provenance coordinates and count per species and provenances out of a total of 81 individual trees.

Provenance	Longitude	Latitude	*A. incana* ($n = 29$)	*B. alleghaniensis* ($n = 20$)	*B. papyrifera* ($n = 7$)	*B. populifolia* ($n = 25$)
Harvard Forest (MA)	-72.2	42.5	7	0	4	7
White Mountains (NH)	-71.0	44.1	6	7	0	9
Dartmouth College (NH)	-70.7	44.9	8	7	2	9
St-Hippolyte (Qc)	-74.0	46.0	8	6	1	0

Supplemental Results

Table S2: Species level slope parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Species	GDD	GSL	SOS	EOS
A. incana	0.07 [-0.04, 0.17]	0.05 [0.00, 0.10]	-0.07 [-0.14, -0.01]	-0.01 [-0.07, 0.04]
B. alleghaniensis	0.13 [0.01, 0.24]	0.03 [-0.02, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
B. papyrifera	0.44 [0.18, 0.70]	0.18 [0.05, 0.32]	0.05 [-0.10, 0.21]	0.21 [0.09, 0.33]
B. populifolia	0.21 [0.07, 0.36]	0.05 [-0.02, 0.13]	0.05 [-0.07, 0.17]	0.10 [0.02, 0.17]

Table S3: Site level intercept parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Site	GDD	GSL	SOS	EOS	Lon.
Harvard Forest (MA)	-0.10 [-0.33, 0.08]	-0.10 [-0.34, 0.08]	-0.05 [-0.26, 0.11]	-0.09 [-0.33, 0.08]	-72.19
White Mountains (NH)	0.08 [-0.09, 0.31]	0.07 [-0.10, 0.29]	0.08 [-0.07, 0.29]	0.08 [-0.10, 0.30]	-71.23
Dartmouth College (NH)	-0.03 [-0.22, 0.15]	-0.05 [-0.26, 0.13]	-0.07 [-0.27, 0.09]	-0.04 [-0.24, 0.15]	-71.15
St-Hippolyte (Qc)	0.04 [-0.14, 0.25]	0.03 [-0.16, 0.24]	0.02 [-0.15, 0.21]	0.05 [-0.14, 0.27]	-74.03

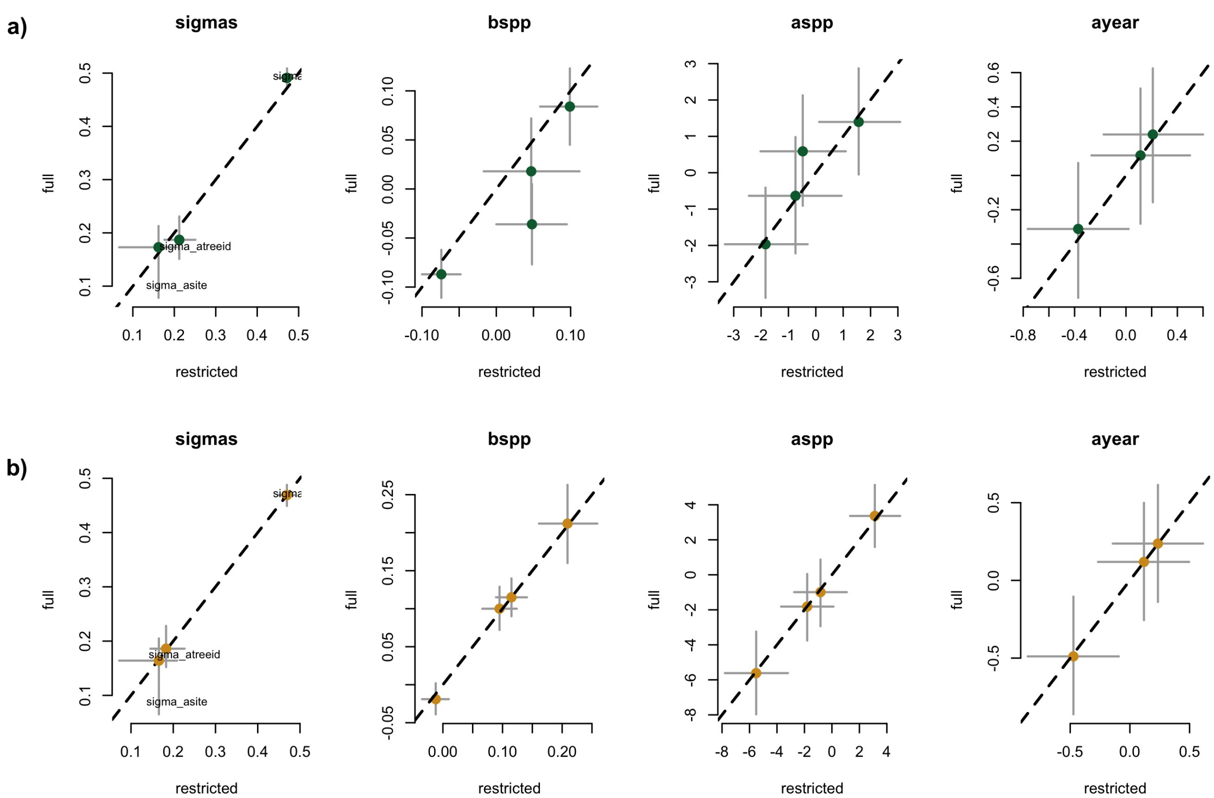

Figure S3: Full vs most restricted rows of data for the model with SOS (a) EOS (b) as the predictor for Wildchrokie. Each pannel represent the different parameters used to fit the models.

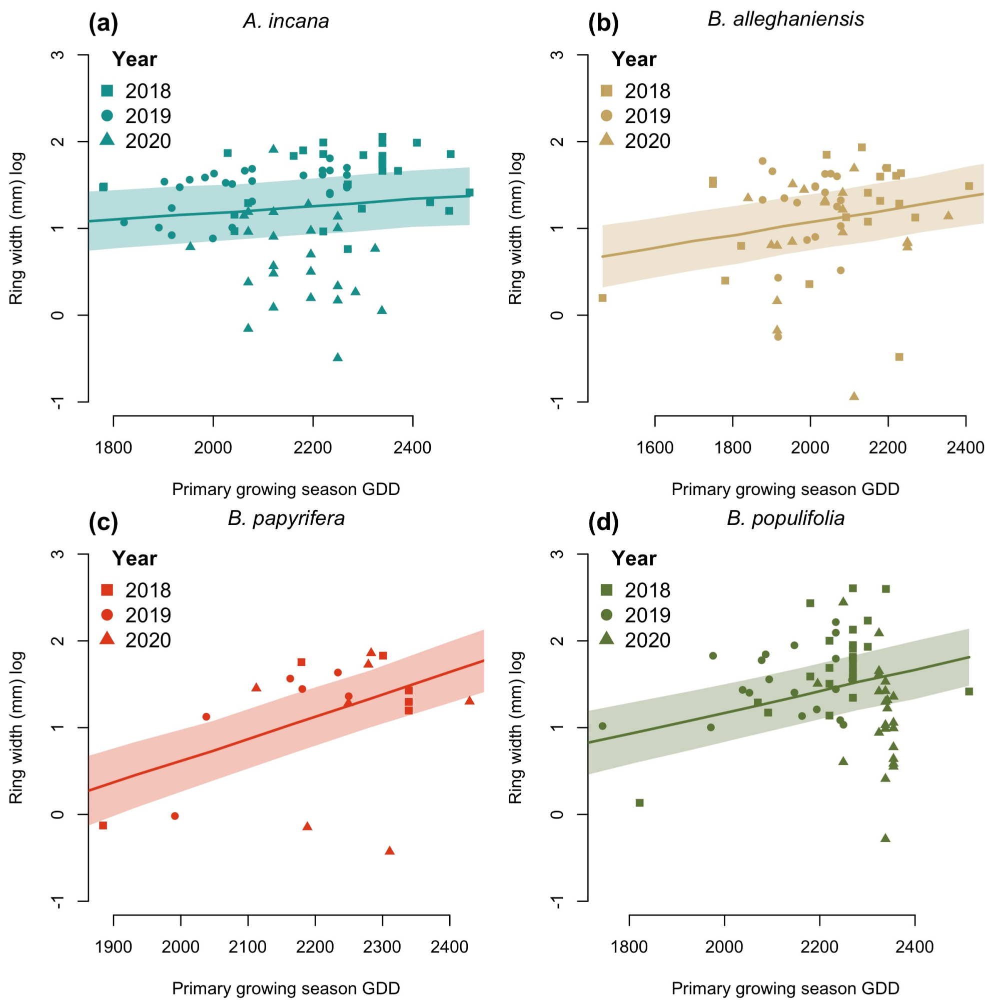

Figure S4: Ring width trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

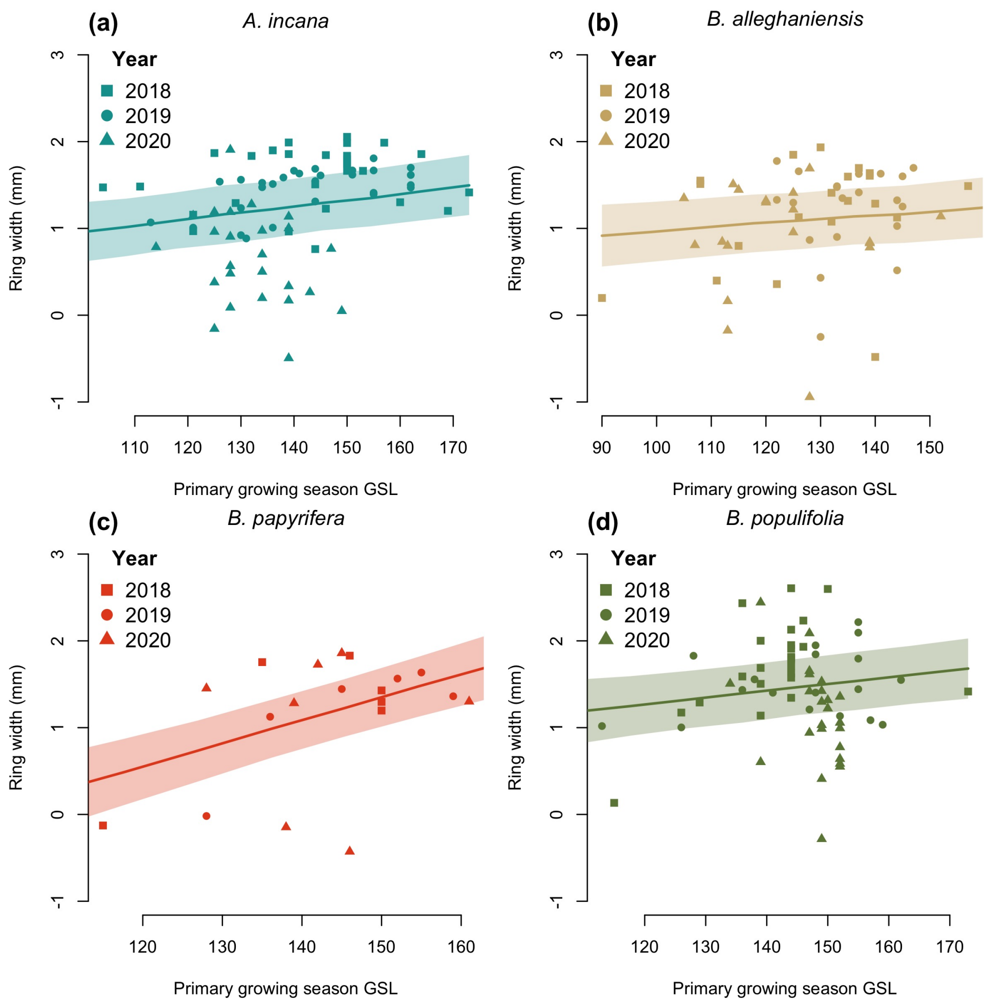

Figure S5: Ring width trends in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

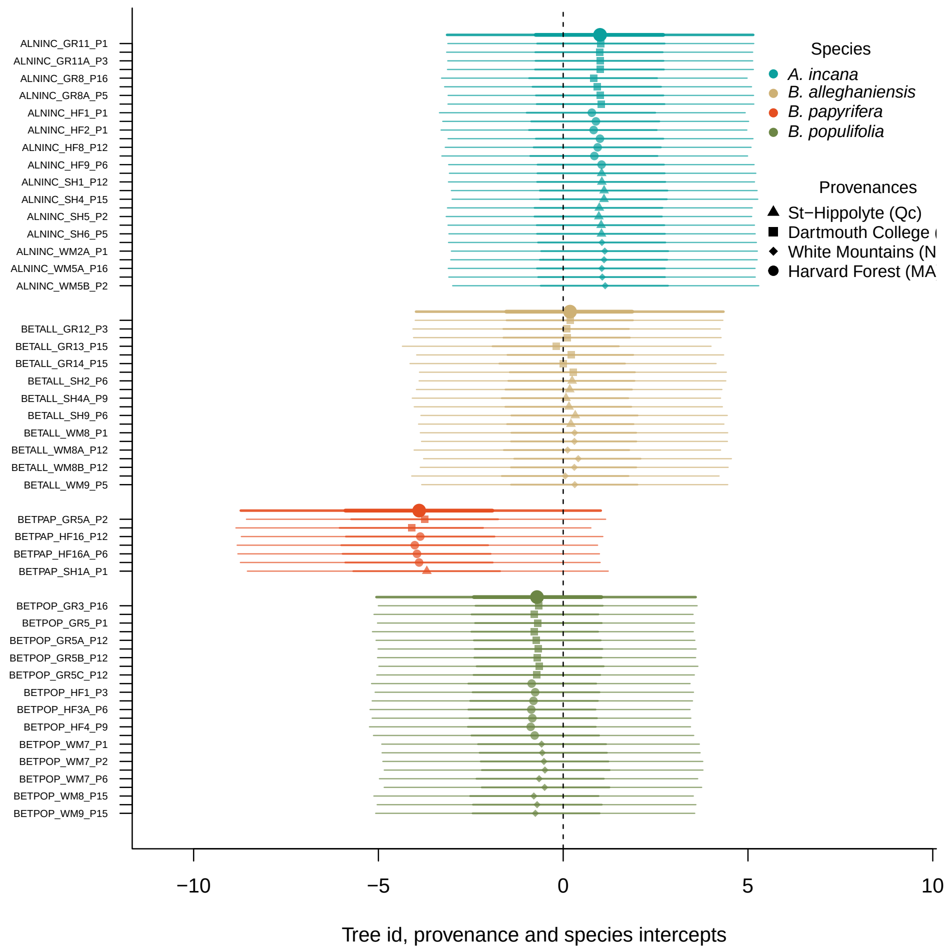

Figure S6: Intercept estimates for each group of the GDD model. The different tree id values represent the combined intercepts of tree id, provenance, species and averaged year intercepts. The bigger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The larger dots and lines represent aggregated average across these values. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

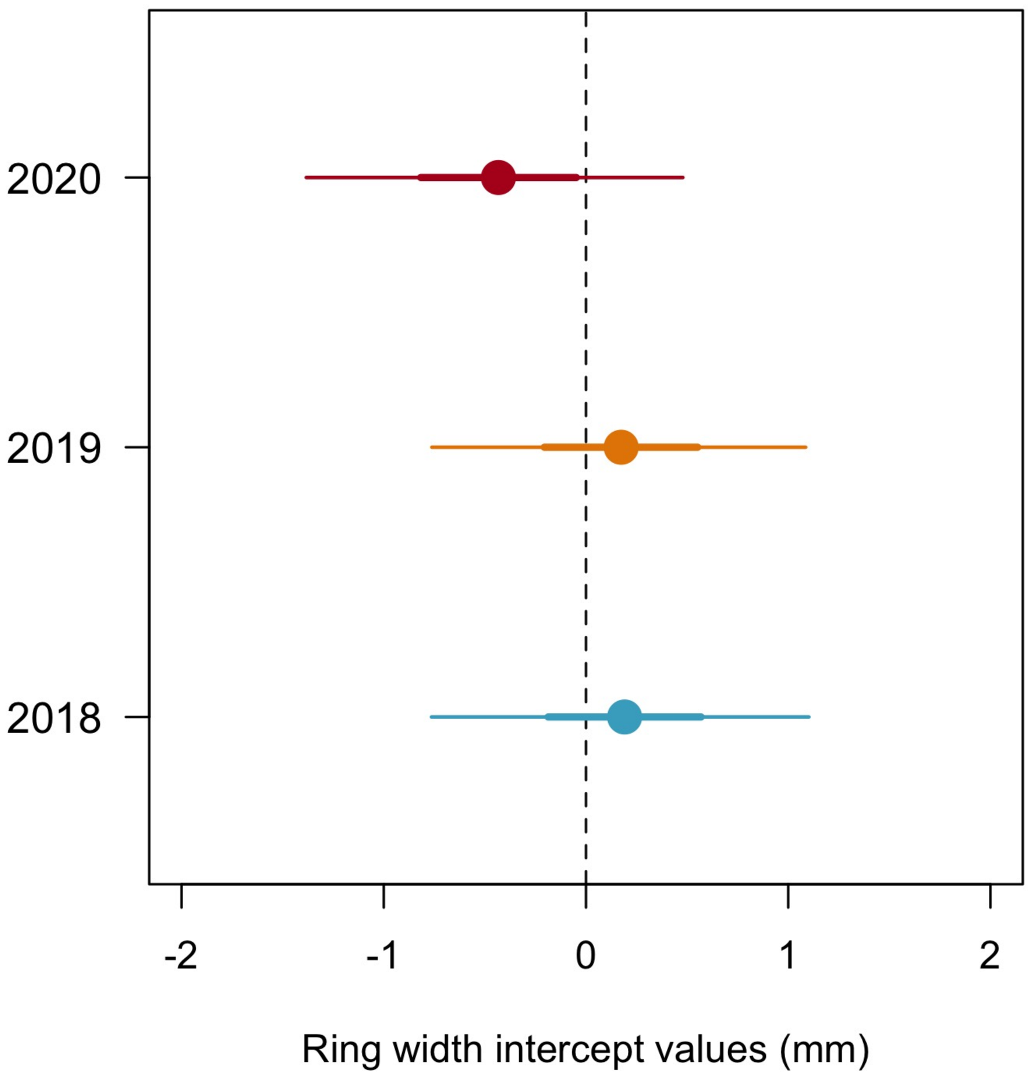

Figure S7: Model estimates for the year intercepts with the dots depicting the mean and the lines, the 50% and 90% uncertainty intervals (UI)

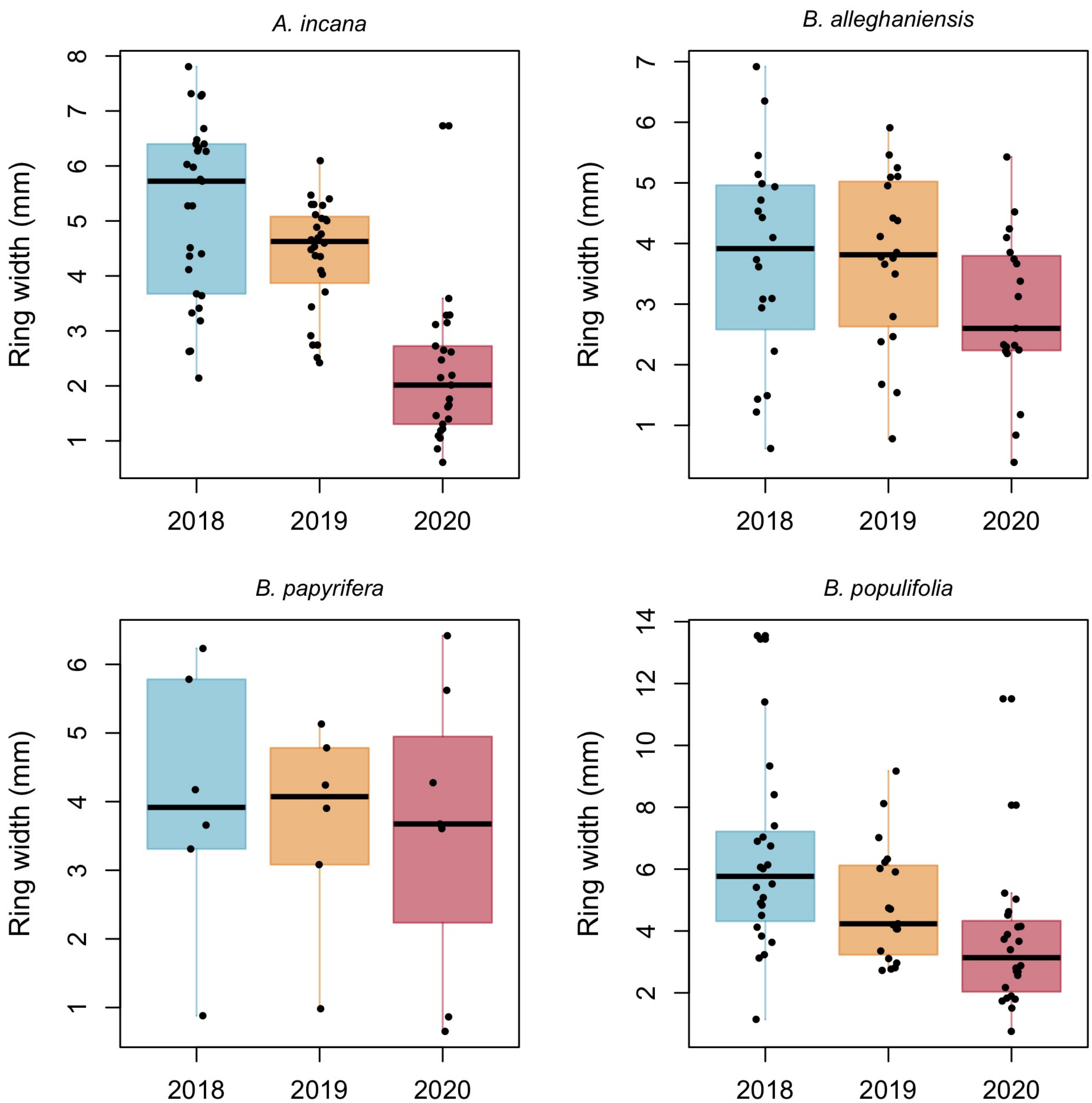

Figure S8: Boxplot representing the empirical data of our ring widths (mm) facetted by species. The dots represent the raw data, the line the mean and the box the IQR.

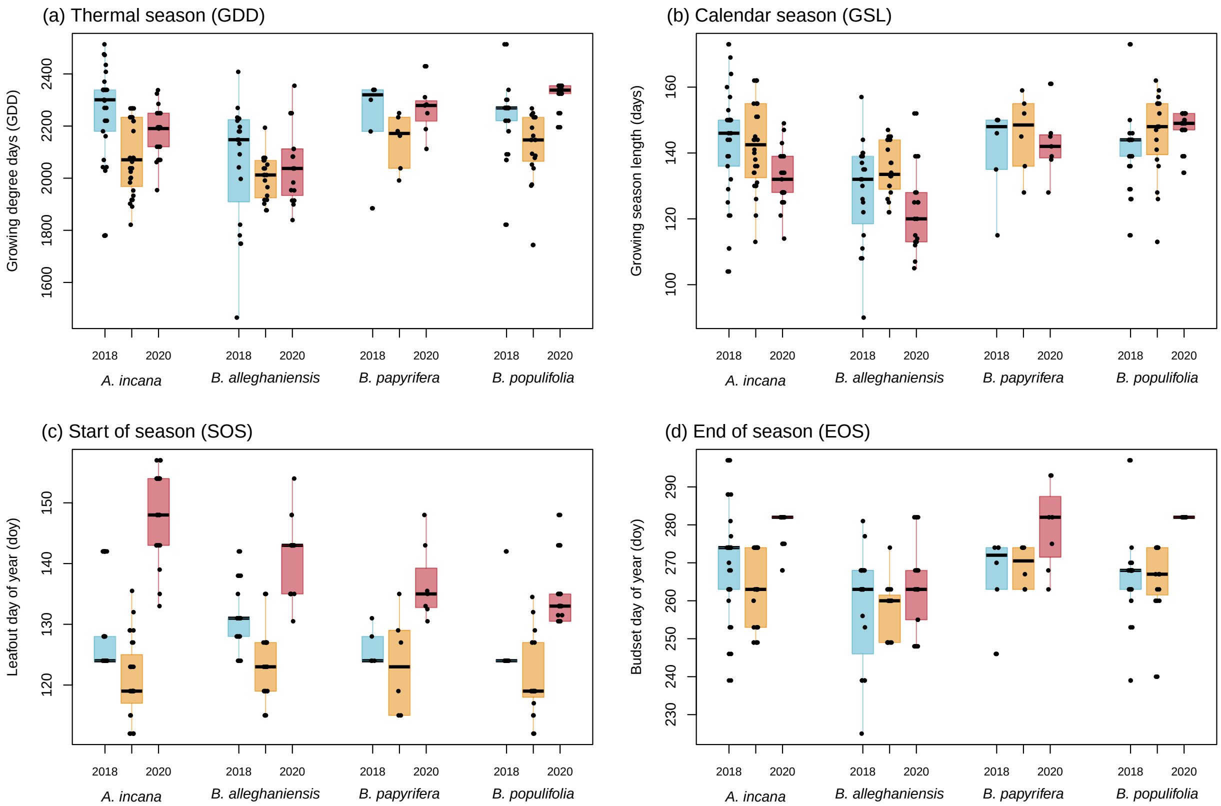

Figure S9: Boxplot representing the empirical data of all out predictors clustered by species, and for each year. The dots represent the raw data, the line the mean and the box the IQR. The year colors match the other year plots (e.g. Figure S7)

Table S4: Posterior means and 90% uncertainty intervals for all model parameters across predictors

Parameter	GDD	GSL	SOS	EOS
$\sigma_{\alpha_{tree}}$	0.17 [0.05, 0.27]	0.18 [0.05, 0.27]	0.21 [0.11, 0.30]	0.18 [0.06, 0.28]
$\sigma_{\alpha_{prov}}$	0.18 [0.02, 0.50]	0.18 [0.02, 0.49]	0.16 [0.01, 0.45]	0.18 [0.03, 0.50]
σ_y	0.48 [0.43, 0.52]	0.48 [0.44, 0.53]	0.47 [0.43, 0.52]	0.47 [0.42, 0.52]
$\beta_{spp_{A.incana}}$	0.07 [-0.04, 0.17]	0.05 [0.00, 0.10]	-0.07 [-0.14, -0.01]	-0.01 [-0.07, 0.04]
$\beta_{spp_{B.alleghaniensis}}$	0.13 [0.01, 0.24]	0.03 [-0.02, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
$\beta_{spp_{B.papyrifera}}$	0.44 [0.18, 0.70]	0.18 [0.05, 0.32]	0.05 [-0.10, 0.21]	0.21 [0.09, 0.33]
$\beta_{spp_{B.populifolia}}$	0.21 [0.07, 0.36]	0.05 [-0.02, 0.13]	0.05 [-0.07, 0.17]	0.10 [0.02, 0.17]
$\alpha_{prov_{DartmouthCollege(NH)}}$	-0.03 [-0.22, 0.15]	-0.05 [-0.26, 0.13]	-0.07 [-0.27, 0.09]	-0.04 [-0.24, 0.15]
$\alpha_{prov_{HarvardForest(MA)}}$	-0.10 [-0.33, 0.08]	-0.10 [-0.34, 0.08]	-0.05 [-0.26, 0.11]	-0.09 [-0.33, 0.08]
$\alpha_{prov_{St-Hippolyte(Qc)}}$	0.04 [-0.14, 0.25]	0.03 [-0.16, 0.24]	0.02 [-0.15, 0.21]	0.05 [-0.14, 0.27]
$\alpha_{prov_{WhiteMountains(NH)}}$	0.08 [-0.09, 0.31]	0.07 [-0.10, 0.29]	0.08 [-0.07, 0.29]	0.08 [-0.10, 0.30]
$\alpha_{year_{2018}}$	0.19 [-0.76, 1.10]	0.20 [-0.77, 1.17]	0.20 [-0.72, 1.14]	0.23 [-0.71, 1.19]
$\alpha_{year_{2019}}$	0.17 [-0.76, 1.09]	0.07 [-0.89, 1.05]	0.10 [-0.83, 1.05]	0.12 [-0.83, 1.09]
$\alpha_{year_{2020}}$	-0.43 [-1.38, 0.48]	-0.38 [-1.35, 0.60]	-0.39 [-1.31, 0.56]	-0.47 [-1.42, 0.50]
$\alpha_{spp_{A.incana}}$	1.01 [-3.14, 5.18]	0.12 [-4.04, 4.32]	1.58 [-1.98, 5.48]	3.08 [-1.29, 7.36]
$\alpha_{spp_{B.alleghaniensis}}$	0.15 [-3.95, 4.36]	0.34 [-3.82, 4.49]	-1.82 [-5.51, 2.13]	-1.82 [-6.29, 2.64]
$\alpha_{spp_{B.papyrifera}}$	-3.82 [-8.73, 1.09]	-2.67 [-7.34, 1.89]	-0.76 [-4.84, 3.59]	-5.58 [-11.03, -0.04]
$\alpha_{spp_{B.populifolia}}$	-0.69 [-4.89, 3.51]	0.22 [-4.00, 4.45]	-0.42 [-4.24, 3.59]	-0.85 [-5.47, 3.82]

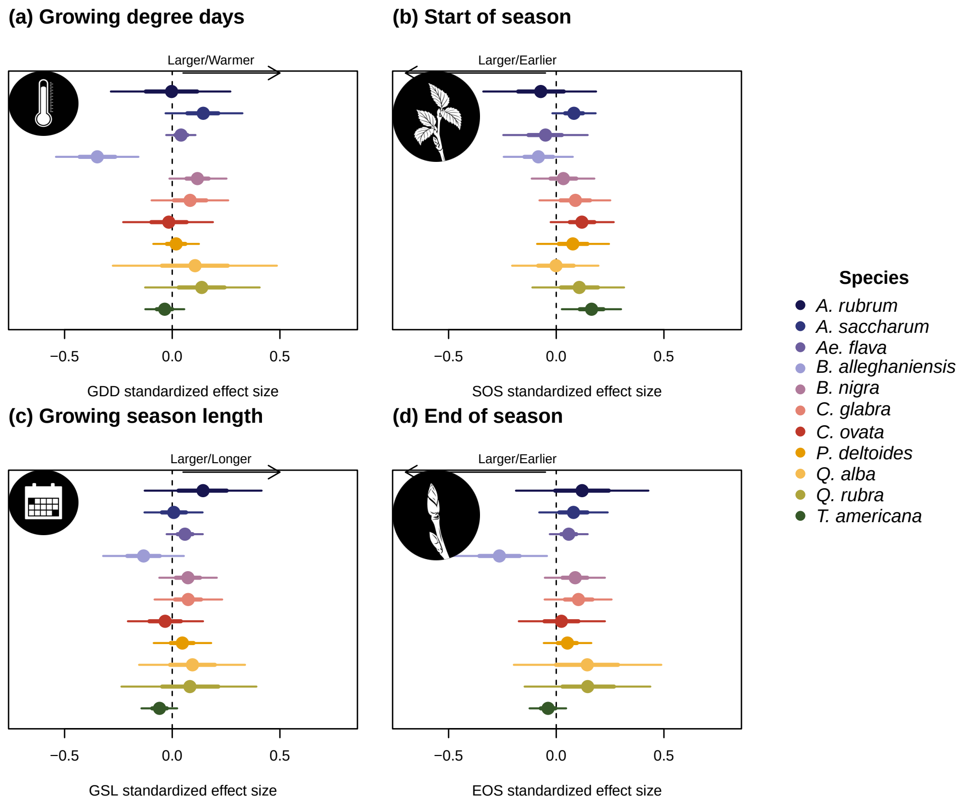

Figure S10: Standardized (z-scored) slope estimates for Wildchrokie. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

Table S5: Posterior means and 90% uncertainty intervals for all model parameters, when using basal area increment (BAI) as the response variable

Parameter	Estimate
$\sigma_{\alpha_{tree}}$	0.52 [0.41, 0.64]
$\sigma_{\alpha_{prov}}$	0.27 [0.03, 0.73]
σ_y	0.63 [0.57, 0.70]
$\beta_{spp_{A.incana}}$	0.11 [-0.03, 0.25]
$\beta_{spp_{B.alleghaniensis}}$	0.20 [0.02, 0.36]
$\beta_{spp_{B.papyrifera}}$	0.92 [0.49, 1.34]
$\beta_{spp_{B.populifolia}}$	0.41 [0.20, 0.61]
$\alpha_{site_{DartmouthCollege(NH)}}$	-0.11 [-0.47, 0.17]
$\alpha_{site_{HarvardForest(MA)}}$	-0.09 [-0.45, 0.19]
$\alpha_{site_{St-Hippolyte(Qc)}}$	0.06 [-0.23, 0.42]
$\alpha_{site_{WhiteMountains(NH)}}$	0.15 [-0.11, 0.52]
$\alpha_{year_{2018}}$	-0.36 [-1.32, 0.61]
$\alpha_{year_{2019}}$	0.33 [-0.65, 1.32]
$\alpha_{year_{2020}}$	-0.04 [-1.01, 0.94]
$\alpha_{spp_{A.incana}}$	3.23 [-2.25, 8.55]
$\alpha_{spp_{B.alleghaniensis}}$	2.09 [-3.40, 7.54]
$\alpha_{spp_{B.papyrifera}}$	-7.59 [-14.84, -0.29]
$\alpha_{spp_{B.populifolia}}$	-0.07 [-5.98, 5.57]